

Ojas Srivastava

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Software Engineer Intern — Monomer Research

B.Tech Artificial Intelligence, SVNIT Surat | **Batch 2028** (Pre-final) | CGPA **9.20/10**
Modern C++ | **Linux** | **DSA** | **OS** | **Computer Architecture** | **Concurrency** | **Networking**

Summary

Pre-final-year **B.Tech** student (Batch **2028**), CGPA **9.20/10**, applying for **Software Engineer Intern** at Monomer Research to build and optimize **high-performance** software in **modern C++** on **Linux**. Primary contest language is **C++** (STL, C++17). Strong foundation in **Data Structures & Algorithms**, **Operating Systems**, and **Computer Organization & Architecture**; project work on **latency**-sensitive paths, **concurrency**, and real-time **networking** (WebSockets/TCP-style streaming). Interested in low-latency infrastructure, reliability under load, and performance engineering—no prior proprietary trading experience required; motivated to learn HFT/MFT systems engineering.

Education

Bachelor of Technology in Artificial Intelligence (CS-related)

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat, India

CGPA: 9.20/10

Aug 2024 – May 2028

- **Eligibility:** Graduating batch **2028**; CGPA **9.20/10** (≥ 7.0); enrolled pre-final year; no active backlogs
- **Coursework:** **Data Structures & Algorithms**, **Operating Systems**, **Computer Organization & Architecture**, OOP (C++), Computer Networks (TCP/IP), DBMS, Distributed Systems, Probability & Statistics

Competitive Programming — Modern C++

- **LeetCode Knight:** Max rating **2048**, **711+** problems, 32 contests — DSA implemented primarily in **C++** (profile)
- **Codeforces Specialist:** Max rating **1419**, **252+** solved — timed contest coding in **C++** (profile)
- Focus: asymptotic complexity, memory/time trade-offs, correct fast implementations under pressure

Projects (Performance / Concurrency / Systems)

LogiFlow — Latency-Aware Large-Scale Pipeline (Google Solution Challenge 2026, Global Top 106)

Apr 2026 – Jun 2026

C++, Python, Redis, SQL, FastAPI, caching, simulation-style validation

GitHub | Live

- Built **high-performance** read paths (**100–400 ms**) across **580+ corridors** / **9,526 stations** by splitting hot-path **Redis** caching from heavy compute—latency vs throughput trade-off thinking.
- Modeled delay risk on **15,650** train-day records (feature design + Gradient Boosting simulator, MAE **22.7 min**); locked correctness with **100/100** automated regression checks—stress-test / simulation mindset before trusting outputs.
- Used **C++** and Python where each fit; owned modular service boundaries so performance changes stay isolatable and measurable.

AirHelp — Concurrent Real-Time Services (PowerMind Hackathon 2026)

2026

Python, FastAPI, WebSockets, NetworkX, A* graph search, Linux deploy

GitHub

- Shipped **concurrent** real-time services over **WebSocket** streams; profiled and tuned end-to-end latency to **1–3s** under **50–100 concurrent** connections.
- Implemented graph-based **A*** navigation (algorithm design + complexity-aware coding) with **95%+** path accuracy on custom walking graphs—DSA applied to a live networked system.

Personal Markets Data Toolkit

2022 – Present

Python, Pandas, NumPy, SQL, REST APIs

Personal

- **3+ years** active Indian **equities** / **mutual funds** investing; scripts for position logging, return attribution, and watchlist screening—practice reading noisy market data and testing ideas with evidence (not production trading infra).

Experience

Software Engineering Intern

IFFCO (Indian Farmers Fertiliser Cooperative), Phulpur Unit, Prayagraj, India

Jun 2025 – Jul 2025

- Designed, developed, tested, and deployed production services on **Linux** (Docker, CI/CD): **10+ REST APIs**, optimized SQL, structured logging, and production debugging.
- Owned reliability through unit/integration tests and code review—correctness and operability under real usage.

Technical Skills

Modern C++ / Linux: C++17, STL, OOP, g++, debugging, Git; develop and run software on **Linux**

Role-critical CS: Data Structures & Algorithms, Operating Systems, Computer Organization & Architecture, complexity analysis

Performance & systems: Latency optimization, caching (Redis), concurrency, real-time networking (WebSockets / TCP-style streams), profiling mindset

Also: Python, C, SQL, Docker, CI/CD, pytest, probability & statistics foundations

Leadership

- **Google Solution Challenge 2026 — Global Top 106** (LogiFlow, Technical Co-Lead)
- **Executive Member, ACM SVNIT & Mentor, Nexus SVNIT** — DSA workshops and contest mentoring in **C++**